

Network goodness calculus proposition

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Part three

The network goodness

A number that measures how the network is good to service the area that it is embedded in. The network length is the sum of all the segments' lengths the network consists of.

Definition 1 (Network length).

For a given network $\bigcup_{j=1}^s \overline{A_j B_j}$, the length is

$$l = \sum_{j=1}^s |A_j B_j| = \sum_{j=1}^s \sqrt{(x_{B_j} - x_{A_j})^2 + (y_{B_j} - y_{A_j})^2}. \quad (0.1)$$

Definition 2 (The goodness function property).

A formula $g(k, M, l)$ defines a network goodness function in three variables properly, iff the function increases by k and decreases by l and M .

The first formula proposal

Proposition 1.

If the number of zones gravitating to the corridor is k , the network inertia is M , and the network's length is l , then the network goodness function is given with the formula:

$$g_1(k, M, l) = \frac{k^2 + k^3}{l^2 + M}. \quad (0.2)$$

Discussion

- proof: $\frac{\partial g_1}{\partial k} \geq 0 \geq \frac{\partial g_1}{\partial l}, \frac{\partial g_1}{\partial M}$.
- by the way: $\frac{\partial g_1(k, M, l)}{\partial k} > 0 > \frac{\partial g_1(k, M, l)}{\partial M} > \frac{\partial g_1(k, M, l)}{\partial l}$.
- denominator reason - adding km^2 with km^2
- nominator - adding dimensions - square l^2 with cube $Q \cdot d_i^2$

The second formula proposal

Proposition 2.

Suppose k , M , and l are as in Proposition 1. Then the network goodness function is given with the formula:

$$g_2(k, M, l) = \frac{k^2}{l^2} + \frac{k^3}{M}. \quad (0.3)$$

Discussion

- proof: $\frac{\partial g_1}{\partial k} \geq 0 \geq \frac{\partial g_1}{\partial l}, \frac{\partial g_1}{\partial M}$.
- no additional discussion

Ideal formula proposal

Proposition 3.

Suppose k, M and l are as in the Propositions above. Let d and D be positive values from the previous.

In particular, n is the number of all demand zones in the environment that the network is set to serve. Then, the network goodness function is given with the formula:

$$g(k, M, l) = \frac{k^2 \cdot \mu(\mathcal{U})}{2 \cdot n^2 \cdot D \cdot l} + \frac{k^3 \cdot d^2 \cdot \sum_{i=1}^k Q(\mathcal{T}_i)}{2 \cdot n^3 \cdot M}. \quad (0.4)$$

Corollary 3.

If $n = k$, and $\mu(\mathcal{U}) = D \cdot l$, and $d_i = d$ for each $\mathcal{T}_i \subset \mathcal{U}$, then the value calculated by formula (0.4) turns to be $1 = 100\%$.

Exponents' discussion

Square zones' number in the first and cube in the second fraction of goodness formula:

- In our patent results $n^2 \geq \mu(\mathcal{U})$, while $n \leq \mu(\mathcal{U})$.
- In our patent results $n^3 \geq \sum_{i=1}^n Q(\mathcal{T}_i)$, while $n^2 \leq \sum_{i=1}^n Q(\mathcal{T}_i)$.

Theorem

Theorem 4.

Beside the values from above, define α and β as follows

$$\alpha = \lceil \log_n \mu(\mathcal{U}) \rceil \text{ and } \beta = \lceil \log_n \sum_{i=1}^n Q(\mathcal{T}_i) \rceil. \quad (0.5)$$

If $\alpha, \beta > 0$, then the value of the function

$$\mathcal{G}(k, M, l) = \frac{k^\alpha \cdot \mu(\mathcal{U})}{2 \cdot n^\alpha \cdot D \cdot l} + \frac{k^\beta \cdot d^2 \cdot \sum_{i=1}^k Q(\mathcal{T}_i)}{2 \cdot n^\beta \cdot M} \quad (0.6)$$

is depending on k increasingly and on l, M decreasingly. Moreover, $\mathcal{G}$ is within $[0, 1]$ depending on d and D , that are given arbitrarily.